S1 - Forecast Uncertainty Derivation

After the float gets its last Gps position, we do not know its exact future position

So,

Time after Gps fix ⟶ position Uncertainty

we Considered the Uncertainty as a diffusion / random - walk process

for a random process

$$\text{Variance} \propto \text{Time}$$

Therefore

$$\sigma^2(t) \propto t$$

and since standard deviation is the square root of variance

$$\sigma(t) \propto \sqrt{t}$$

At $t=0$, the float already has some Gps/fix Uncertain

$$\sigma(0) = \sigma_0$$

We also introduced τ, the Current-error decorrelation time to represent how quickly the Current error changes

$$\frac{t}{\tau}$$

$$\boxed{\sigma(t) = \sigma_0 \sqrt{1 + \frac{t}{\tau}}}$$

I/p

σ_0 → uncertainty Immediately after Gps fix

t → time since last Gps fix

τ → Current error decorrelation time

o/p

$\sigma(t)$ = predicted position uncertainty at time t_0

S4 - up - Current Loiter point Derivation

The float is moving with the ocean current.
Suppose the float is predicted to surface at x_f
But the glider reaches the waiting location before the
Float surfaces
so if the glider simply waits at x_f, the float
move down stream before the actual meeting

Calculate how far the float moves

Current velocity u

waiting time t_{wait}

Basic motion equation

Distance = Velocity × Time

$$\therefore d_{drift} = u t_{wait}$$

Because the float will move in the direction of the current, the glider should wait opposite to the current direction

$$\therefore x_{loiter} = x_f - u t_{wait}$$

final formula is

$$x_{loiter} = x_f - u t_{wait}$$

Float energy

Inputs

$x_f \rightarrow$ predicted float surfacing position

$u \rightarrow$ ocean - current velocity

$t_{wait} \rightarrow$ expected waiting / transmission time

output

x_{loiter} = position where the glider should wait

S7 - polar Solar energy Derivation

At polar latitudes the amount of sunlight changes dramatically with the season

first Calculate Solar declination

we use $\delta = 23.44^\circ \cos\left[\frac{360^\circ}{365}(N+10)\right]$

N = day of the year

δ = Solar declination

Calculate the Sunset hour angle

At latitude ϕ:

$\cos\omega = -\tan\phi \tan\delta$

for Therefor

$\boxed{\omega = \cos^{-1}(-\tan\phi \tan\delta)}$

for the project

$\boxed{\phi = -62^\circ}$

Convert angle into daylight hours

earth rotates

15° / hour

$\therefore D = 200/15^\circ$

compare energy generated with energy consume

System load :

E_{load}

$\therefore NER = \frac{E_{harvest}}{E_{load}}$

NER = Net energy Ratio

if $\boxed{NER \geq 1}$

then

Energy generated $\geq$ energy consumed

→ system can sustain itself

if $\boxed{NER < 1}$

then

Energy generated < energy consumed

→ battery reserves are required..